

REDACTA AI

Your PII never leaves the machine.

On-device detection & redaction for Indian documents.

Snapdragon® AI Lab Build & Present Challenge · Sudhanshu Upadhyay · github.com/upadhyayaditya7/redacta-ai

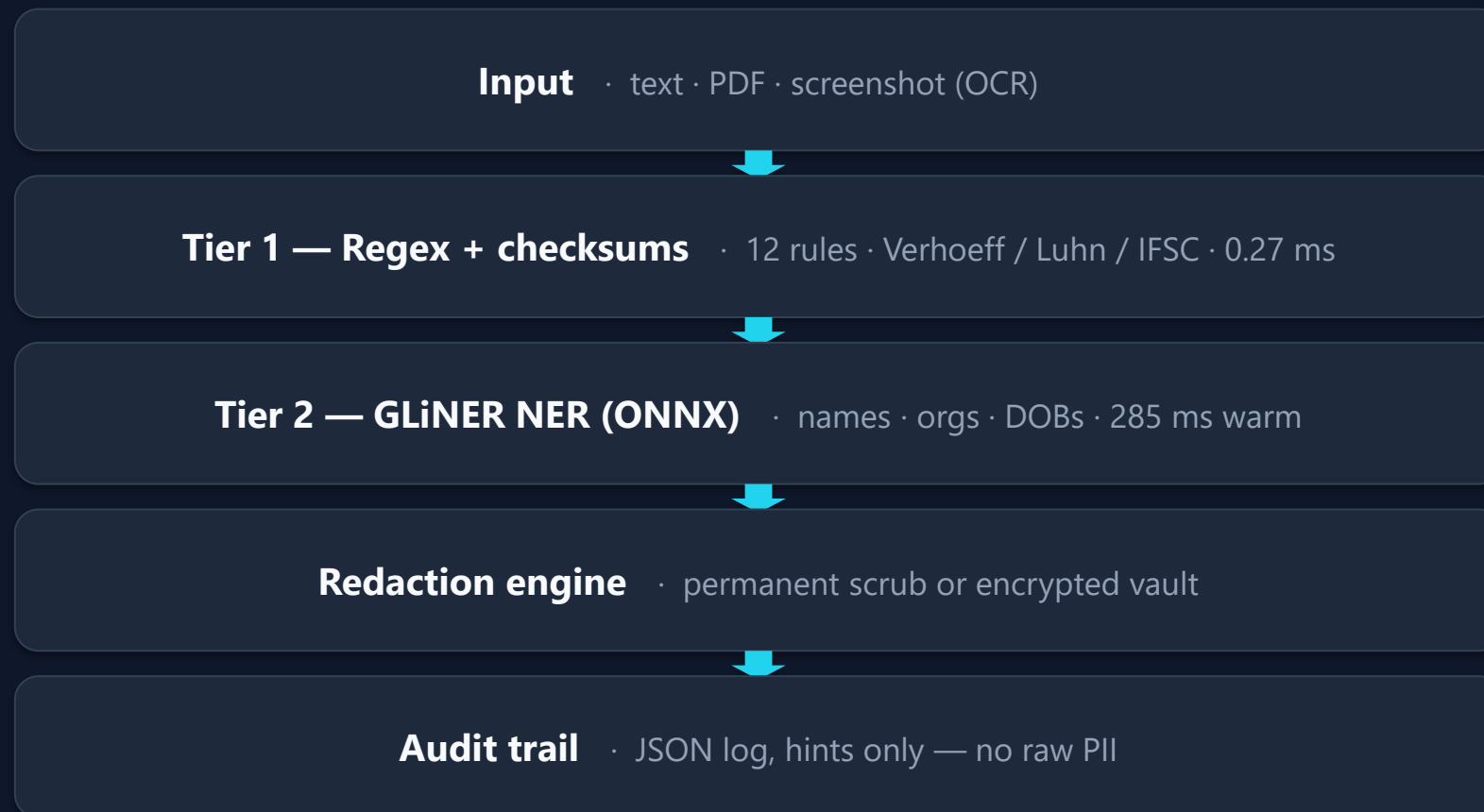
The safest document is the one you never upload

- **Cloud redaction is an oxymoron** — you send the most sensitive files you own to a third party to make them shareable.
- **Global tools miss India.** Microsoft Presidio scores F1 0.373 on Indian financial PII — it was never built for Aadhaar, IFSC or UPI.
- **Regex alone is not enough.** Names, organisations and dates of birth have no pattern — but a 12-digit order ID must not be flagged as an Aadhaar number either.
- **Redaction is usually a one-way door.** Teams need shareable copies and the original; permanent deletion breaks the workflow.

Drop a document. Get redactions you can undo.

- **Detect with proof.** Every hit carries a score and a human-readable reason: Verhoeff checksum valid (UIDAI), Luhn valid, PAN holder-type letter valid.
- **Redact two ways.** Permanent scrub, or reversible «RDCT-...» tokens whose originals sit in a local encrypted vault (PBKDF2-HMAC-SHA256, 200k iterations + HMAC integrity).
- **Text, PDF, screenshots.** One CLI and one Streamlit UI; batch mode watches a folder and redacts arrivals drop-box style.
- **Compliance-safe audit trail.** The JSON audit log records what was found and what was done — with hints, never raw values.

Two tiers: a deterministic floor, an AI ceiling



Graceful by design

No torch installed? Regex tier still runs.

Model missing? Degrades and says so honestly.

Vault lost the passphrase? That's the point — irreversible to everyone, including us.

2.4× Microsoft Presidio on Indian financial PII

System	Precision	Recall	F1
Redacta Full (regex + GLiNER)	1.000	0.831	0.907
Redacta Regex only	0.900	0.692	0.783
Microsoft Presidio	0.975	0.231	0.373

- **Zero false positives on the eval set** (P = 1.000) — every flagged span explains itself, so a human can verify in seconds.
- **Fully reproducible:** `python tests/test_accuracy.py` regenerates the eval on 10 labelled synthetic documents.

Compiled and executed on the real X Elite NPU

- **Pipeline:** GLiNER-P11 → ONNX (static shapes) → QNN context binary → Snapdragon X Elite CRD (Hexagon v73) via AI Hub.
- **Real jobs, real device:** compile jgnz1qovg SUCCESS · on-device inference jp1no1r8g SUCCESS with genuine tokenized input.
- **Two upstream fixes contributed:** int64 ReduceMax lowered to fp32 casts; span-Einsum rewritten as Transpose+MatMul — proven bitwise-identical (9/9 entities, max diff 0.0).
- **On-device load:** 17.7 s cold → 1.2 s warm; every load/inference step logged by the device runtime.

Honest benchmarks beat optimistic ones

0.959

logits cosine, HTP fp16 vs CPU fp32

0 / 9

entities surviving fp16 execution

0 / 9

entities surviving naive int8 (CPU)

- **The finding:** GLiNER's LSTM span-scorer runs on razor-thin logit margins; fp16 accumulation across 4,950 nodes compresses the range (CPU [-68.6, +6.6] → HTP [-15.6, -1.5]) and erases the head.
- **The engineering answer:** the two-tier design absorbs it — the product ships the CPU AI tier (285 ms) and stays correct; every number is in benchmarks/npu.md \$2b with job IDs.
- **The roadmap:** an NPU-viable span head (DeBERTa-style) and vocabulary pruning (the embedding table is 66% of the artifact) are the two moves queued next.

Runs anywhere a pip install works

- **Zero special hardware needed to run:** CPU-only ONNX Runtime, no torch at inference; the regex tier runs with the AI stack absent entirely.
- **Built on a conventional PC, validated on the target:** development happened on an Intel machine; NPU validation used AI Hub's cloud Snapdragon X Elite CRD — the exact hardware the challenge targets.
- **Two front doors:** `python app.py ...` CLI for scripts and batch folders, Streamlit UI with file upload + download buttons for everyone else.
- **38 tests, every benchmark reproducible** with one command; sample data is synthetic — no real identifiers anywhere.

Next 90 days

- **NPU-viable detector head.** Swap the LSTM span-scorer for a DeBERTa-style head that survives fp16, re-export, re-verify on the same AI Hub pipeline.
- **Vocabulary pruning.** 250k multilingual tokens → the subset reachable from Indian financial documents; 1.1 GB artifact → ~400 MB, no accuracy cost on this domain.
- **Calibrated quantisation.** QNN PTQ with calibration data instead of CPU dynamic int8 — the path npu.md already points to.
- **Watch-mode daemon + policy packs.** Folder policies per department (finance, HR) with separate vaults.

Nothing leaves the machine.

Every claim in this deck is a command you can run.

github.com/upadhyayaditya7/redacta-ai · 38 tests · benchmarks/npu.md · docs/Redacta_AI_Proposal.pdf